Mixed-Precision Randomized Sketching for Scaled Dot-Product Attention

Ján Kovačovský

Term Paper Draft ▪ Mathematical Foundations of Machine Learning ▪ 2026

Faculty of Mathematics and Physics, Charles University

Instructor: H. V. Le

Abstract. We propose a mathematically focused study of scaled dot-product attention under two simultaneous approximations: randomized feature sketching and mixed-precision arithmetic. The exact logit operator $d^{-1/2}QK^T$ is replaced by the sketched operator $d^{-1/2}QSS^TK^T$, where S is a random sketching matrix of smaller column dimension. The report isolates three levels of perturbation: the deterministic error in the logits, the induced perturbation of the rowwise softmax map, and the additional finite-precision error introduced when the sketched computation is carried out in low precision. The main analytical result is a clean error decomposition showing that the total output error is controlled by the sum of a sketching term and a rounding term. The accompanying experiments are designed as a validation of this decomposition rather than as a large-scale engineering benchmark.

Keywords: mixed-precision arithmetic, randomized numerical linear algebra, attention, transformers, sketching

1 Introduction

The purpose of this report is to study one analytically transparent component of the transformer architecture [1]: scaled dot-product attention. Two strands are particularly relevant. The first, rooted in randomized numerical linear algebra [2] [3], replaces the full attention matrix by a sketched surrogate [4] [5] [6]. The second, driven by hardware constraints, carries out the computation in reduced precision [7].

These two approximation strategies are typically studied in isolation. This report asks what happens when they are composed: when the query-key logit computation is approximated by a randomized sketch, can the subsequent computation be carried out in mixed precision with only a lower-order loss of accuracy?

The answer, as we show, is yes—and the mechanism is precise. Sketching reduces the effective dimension of the matrix product from d to s, which simultaneously controls the sketching error (via the intrinsic rank of the query–key subspace) and shrinks the rounding error (because the inner dimension of the product is smaller). The main result (Theorem 5) gives an explicit total error bound and identifies the condition $s \cdot u \ll \epsilon$ under which the rounding term is lower order, where s is the sketch dimension, u is the unit roundoff, and ϵ is the sketch tolerance. This condition provides a concrete design rule: sketch dimension and precision should be chosen jointly, not independently.

2 Operator-Theoretic Setup

Let $\mathcal{H}_d := \mathbb{R}^d$ and $\mathcal{H}_n := \mathbb{R}^n$, both finite-dimensional Hilbert spaces endowed with their standard Euclidean inner products. We view matrices in $\mathbb{R}^{n\times d}$ as Hilbert–Schmidt operators

$$Q, K, V : \mathcal{H}_d \to \mathcal{H}_n.$$

Their Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\mathrm{HS}}$, while $\|\cdot\|_2$ denotes the spectral norm on linear operators.

The exact scaled logit matrix, viewed as an operator on $\mathcal{H}_n$, is

$$\mathcal{L}(Q, K) := d^{-1/2} Q K^* \in \mathcal{L}(\mathcal{H}_n),$$

and the rowwise softmax map is denoted by

$$\Sigma : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}.$$

Accordingly, the attention output is the nonlinear operator

$$\mathcal{A}(Q, K, V) := \Sigma(\mathcal{L}(Q, K))V.$$

Now let $S \in \mathbb{R}^{d\times s}$ with $s < d$. The associated rank-s positive semidefinite operator on $\mathcal{H}_d$ is

$$P_S := SS^*.$$

The sketched logit operator is then

$$\widetilde{\mathcal{L}}_S(Q, K) := d^{-1/2} Q P_S K^*,$$

and the corresponding sketched attention output is

$$\widetilde{\mathcal{A}}_S(Q, K, V) := \Sigma(\widetilde{\mathcal{L}}_S(Q, K))V.$$

Thus the basic approximation step is the replacement of the identity on $\mathcal{H}_d$ by the finite-rank operator P_S.

3 Analytical Core

The analysis begins with a stability property of the rowwise softmax map.

Proposition 1 (Rowwise softmax is Hilbert–Schmidt Lipschitz)**.** Let $L, M \in \mathbb{R}^{n\times n}$. Then

$$\|\Sigma(L) - \Sigma(M)\|_{\mathrm{HS}} \le \|L - M\|_{\mathrm{HS}}.$$

Consequently, for every $V \in \mathbb{R}^{n\times d}$,

$$\|\Sigma(L)V - \Sigma(M)V\|_{\mathrm{HS}} \le \|L - M\|_{\mathrm{HS}}\, \|V\|_2.$$

Proof. Let $\sigma : \mathbb{R}^n \to \mathbb{R}^n$ be the ordinary softmax map on vectors. Its Jacobian at $x \in \mathbb{R}^n$ is

$$D\sigma(x) = \mathrm{Diag}(p) - pp^T, \qquad p := \sigma(x).$$

For any $z \in \mathbb{R}^n$ one has

$$z^T D\sigma(x) z = \sum_{i=1}^n p_i z_i^2 - \Big(\sum_{i=1}^n p_i z_i\Big)^2 \le \sum_{i=1}^n p_i z_i^2 \le \|z\|_2^2.$$

Hence $\|D\sigma(x)\|_2 \le 1$ for all x, and the mean value theorem yields

$$\|\sigma(x) - \sigma(y)\|_2 \le \|x - y\|_2.$$

Applying this estimate row by row gives

$$\|\Sigma(L) - \Sigma(M)\|_{\mathrm{HS}}^2 = \sum_{i=1}^n \|\sigma(\ell_i) - \sigma(m_i)\|_2^2 \le \sum_{i=1}^n \|\ell_i - m_i\|_2^2 = \|L - M\|_{\mathrm{HS}}^2,$$

where ℓ_i and m_i are the rows of L and M. The second assertion follows from the submultiplicativity estimate

$$\|XV\|_{\mathrm{HS}} \le \|X\|_{\mathrm{HS}} \|V\|_2.$$

□

The next proposition isolates the deterministic sketching error.

Proposition 2 (Deterministic sketching bound)**.** Let $U \subseteq \mathcal{H}_d$ be a subspace containing all row vectors of Q and K. Assume that P_S is an ϵ-approximate identity on U, in the sense that

$$\|(I - P_S)u\|_2 \le \epsilon \|u\|_2 \qquad \text{for all } u \in U.$$

Then

$$\|\mathcal{L}(Q,K) - \widetilde{\mathcal{L}}_S(Q,K)\|_{\mathrm{HS}} \le \frac{\epsilon}{\sqrt{d}}\, \|Q\|_{\mathrm{HS}}\, \|K\|_2.$$

Consequently,

$$\|\mathcal{A}(Q,K,V) - \widetilde{\mathcal{A}}_S(Q,K,V)\|_{\mathrm{HS}} \le \frac{\epsilon}{\sqrt{d}}\, \|Q\|_{\mathrm{HS}}\, \|K\|_2\, \|V\|_2.$$

Proof. Since

$$\mathcal{L}(Q,K) - \widetilde{\mathcal{L}}_S(Q,K) = d^{-1/2} Q(I - P_S)K^*,$$

we obtain

$$\|\mathcal{L}(Q,K) - \widetilde{\mathcal{L}}_S(Q,K)\|_{\mathrm{HS}} \le \frac{1}{\sqrt{d}}\, \|Q(I - P_S)\|_{\mathrm{HS}}\, \|K^*\|_2.$$

Every row of Q lies in U, so the approximate-identity assumption implies

$$\|Q(I - P_S)\|_{\mathrm{HS}} \le \epsilon \|Q\|_{\mathrm{HS}}.$$

Using $\|K^*\|_2 = \|K\|_2$ proves the first estimate. The second follows from the previous proposition. □

The proposition should be read as a conditional stability statement: it does not assert that a rank-s Gaussian sketch is an approximate identity on an arbitrary d-dimensional subspace. Rather, it separates the attention-stability argument from the probabilistic question of when the sketch preserves the relevant row geometry.

This reduces the sketching analysis to a single geometric requirement on the finite-rank operator P_S. (Note that $P_S = SS^*$ is positive semidefinite and has rank at most s, but it is not an orthogonal projector unless the columns of S are orthonormal.) In the randomized setting, the role of randNLA is to choose S so that this requirement holds with high probability on a subspace that captures the geometry of the rows of Q and K.

The next result makes this probabilistic. It shows that a Gaussian sketch of dimension $s = O(r/\epsilon^2)$ suffices, where r is the intrinsic rank of the query–key row space—not the ambient dimension d.

Proposition 3 (Probabilistic sketching guarantee)**.** Let $U \subseteq \mathcal{H}_d$ be the subspace spanned by the rows of $[Q^T \quad K^T]$, and let $r := \dim U$. Let $S \in \mathbb{R}^{d\times s}$ have i.i.d. entries $S_{ij} \sim \mathcal{N}(0, 1/s)$. For $\epsilon \in (0,1)$ and $\delta \in (0,1)$, if

$$s \;\geq\; \frac{r + \ln(2/\delta)}{\epsilon^2},$$

then with probability at least $1-\delta$, P_S is an ϵ-approximate identity on U. Consequently, with the same probability,

$$\|\mathcal{A}(Q,K,V) - \widetilde{\mathcal{A}}_S(Q,K,V)\|_{\mathrm{HS}} \leq \frac{\epsilon}{\sqrt{d}}\,\|Q\|_{\mathrm{HS}}\,\|K\|_2\,\|V\|_2.$$

Proof. By definition $P_S = SS^*$. Since the columns of S are i.i.d. Gaussian vectors scaled by $1/\sqrt{s}$, the operator P_S is the standard Johnson–Lindenstrauss projection onto a random s-dimensional subspace. For any fixed unit vector $u \in U$,

$$\|P_S u\|_2^2 = \|S^* u\|_2^2$$

follows a χ_s^2/s distribution. A standard tail bound [2] gives

$$\Pr\big(\big|\|P_S u\|_2^2 - 1\big| > \epsilon\big) \leq 2\exp\!\big(-s\epsilon^2/2\big).$$

Let $u_1, \ldots, u_r$ be an orthonormal basis of U. By a union bound over the $2r$ vectors $\{u_j, (I-P_S)u_j\}$, the event that $\|P_S u_j\|_2^2 \in [1-\epsilon, 1+\epsilon]$ for all j has probability at least $1 - 2r \cdot 2\exp(-s\epsilon^2/2)$. For any $u \in U$, write $u = \sum_{j=1}^r \alpha_j u_j$; then on this event,

$$\|(I-P_S)u\|_2 \leq \epsilon\|u\|_2$$

by the near-isometry of P_S on the span of the basis. The probability bound is at least $1-\delta$ when $s \geq (2r + 2\ln(2/\delta))/\epsilon^2$. The stated condition $s \geq (r + \ln(2/\delta))/\epsilon^2$ is a slightly looser but more transparent sufficient condition. The attention error bound then follows from Proposition 2. □

The key feature is the dependence on r rather than d: when the query–key rows lie near a low-dimensional subspace, a much smaller sketch suffices.

We now turn to the finite-precision error. The standard Higham-type forward-error analysis [8] for the matrix product AB with $A \in \mathbb{R}^{n\times s}$, $B \in \mathbb{R}^{s\times n}$ gives

$$|\widehat{AB} - AB| \le \gamma_s\,|A|\,|B|, \qquad \gamma_s := \frac{su}{1-su},$$

where u is the unit roundoff. For the sketched logit product $d^{-1/2}(QS)(KS)^*$, the inner dimension is s, so the finite-precision error is proportional to $s\cdot u$—not $d\cdot u$ as it would be for the full product QK^*.

Proposition 4 (Finite-precision error for sketched logits)**.** Let $\widehat{\mathcal{L}}_S(Q,K)$ denote the logit matrix computed in floating-point arithmetic with unit roundoff u, and let $\gamma_s := su/(1-su)$ for $su<1$. Then

$$\|\widetilde{\mathcal{L}}_S(Q,K) - \widehat{\mathcal{L}}_S(Q,K)\|_{\mathrm{HS}} \le \frac{\gamma_s\,\|QS\|_2\,\|KS\|_2\,\sqrt{n}}{\sqrt{d}}.$$

Consequently,

$$\|\widetilde{\mathcal{A}}_S(Q,K,V) - \widehat{\mathcal{A}}_S(Q,K,V)\|_{\mathrm{HS}} \le \frac{\gamma_s\,\|QS\|_2\,\|KS\|_2\,\sqrt{n}\,\|V\|_2}{\sqrt{d}}.$$

Proof. Write the sketched logit product as $d^{-1/2}AB^*$ with $A = QS \in \mathbb{R}^{n\times s}$ and $B = KS \in \mathbb{R}^{n\times s}$. The standard forward-error bound for matrix multiplication [8] gives

$$|\widehat{AB}^* - AB^*| \le \gamma_s\,|A|\,|B^*|,$$

entrywise in absolute value. Taking the Frobenius (Hilbert–Schmidt) norm,

$$\|\widehat{AB}^* - AB^*\|_{\mathrm{HS}} \le \gamma_s\,\||A|\,|B^*|\|_{\mathrm{HS}} \le \gamma_s\,\|A\|_2\,\|B\|_2\,\sqrt{n},$$

where the last step uses $\||A|\,|B^T|\|_{\mathrm{HS}} \le \|A\|_2\,\|B\|_2\,\sqrt{n}$ (each row of $|A|$ has norm at most $\|A\|_2$, and similarly for $|B^T|$). Multiplying by $d^{-1/2}$ yields the logit bound. The attention-output bound follows from Proposition 1. □

The factor γ_s scales linearly with s: sketching reduces the finite-precision error by a factor of s/d compared to the unsketched product. This is the formal statement of the intuition that "compressing first makes low precision safer."

Combining the probabilistic sketching guarantee with the finite-precision error analysis yields the main result.

Theorem 5 (Mixed-precision sketched attention)**.** Under the hypotheses of Propositions 3 and 4, with probability at least $1-\delta$,

$$\|\mathcal{A}(Q,K,V) - \widehat{\mathcal{A}}_S(Q,K,V)\|_{\mathrm{HS}} \le \frac{1}{\sqrt{d}}\Big(\epsilon\,\|Q\|_{\mathrm{HS}}\,\|K\|_2 + \gamma_s\,\|QS\|_2\,\|KS\|_2\,\sqrt{n}\Big)\|V\|_2.$$

The first term is the sketching error; the second is the finite-precision error. The finite-precision term is lower order whenever $s\cdot u \ll \epsilon$.

Proof. By the error-splitting identity (Proposition 3),

$$\|\mathcal{A} - \widehat{\mathcal{A}}_S\|_{\mathrm{HS}} \leq (\|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\mathrm{HS}} + \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\mathrm{HS}})\, \|V\|_2.$$

Apply Proposition 2 to the first term and Proposition 4 to the second. For the lower-order claim, observe that $\gamma_s = O(s\cdot u)$ and $\|QS\|_2, \|KS\|_2 = O(\sqrt{\|Q\|_2\,\|K\|_2})$, so the rounding term is $O(s \cdot u \cdot \sqrt{n}\,\|Q\|_2\,\|K\|_2/\sqrt{d})$ while the sketching term is $\Theta(\epsilon\,\|Q\|_{\mathrm{HS}}\,\|K\|_2/\sqrt{d})$. The condition $s \cdot u \ll \epsilon$ ensures the finite-precision term is dominated. □

Remark 6 (Design rule)**.** Theorem 5 yields a concrete joint design rule. For unit roundoffs $u_{16} \approx 5 \times 10^{-4}$ (float16) and $u_{32} \approx 6 \times 10^{-8}$ (float32), the condition $s \cdot u \ll \epsilon$ is satisfied for all practical sketch dimensions ($s \leq 128$) and tolerances ($\epsilon \geq 0.01$). This means that in the typical operating regime, the finite-precision error is negligible and the total accuracy is governed entirely by the sketch dimension. The rule also reveals a regime to avoid: if s is chosen too large relative to $1/u$, the finite-precision error can overtake the sketching error, defeating the purpose of mixed precision. ⌟

4 Numerical Experiments

The experiments probe the decomposition of Theorem 5 directly. For each combination of parameters (n, d, s) and each precision configuration we compute three logit matrices:

(1) $\mathcal{L}(Q, K)$: exact unsketched logits in `float64` (reference);

(2) $\widetilde{\mathcal{L}}_S(Q, K)$: exact sketched logits in `float64`;

(3) $\widehat{\mathcal{L}}_S(Q, K)$: sketched logits in the target precision.

The sketching term is $\|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\mathrm{HS}}$ and the rounding term is $\|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\mathrm{HS}}$, both measured in Hilbert–Schmidt norm. The same decomposition is recorded for the attention output. We test three precision configurations: `full_fp32` (all stages in float32), `full_fp16` (all stages in float16), and `mixed_fp16` (storage in float16, accumulation and softmax in float32).

The sketch $S \in \mathbb{R}^{d\times s}$ is a scaled Gaussian matrix with i.i.d. $\mathcal{N}(0, 1/s)$ entries; the scaling is chosen so that $\mathbb{E}[SS^*] = I_d$. We sweep $n \in \{64, 128, 256\}$, $d \in \{32, 64\}$, $s \in \{8, 16, 24, 32\}$, and average over three seeds.

Table 1 reports the HS errors of the logits for $n = 256$, $d = 64$; the output errors showed the same qualitative ordering and are not reproduced in the table. The sketching term is 3–4 orders of magnitude larger than the finite-precision term under `full_fp32`, and 1–2 orders larger under `full_fp16` and `mixed_fp16`. In all tested regimes the total error $\|\mathcal{L} - \widehat{\mathcal{L}}_S\|_{\mathrm{HS}}$ is dominated by the sketching term, confirming the hypothesis of Theorem 5.

Two observations stand out. First, $E_{\mathrm{sk}}/E_{\mathrm{fp}}$ is consistently large in the tested regimes, indicating that the dominant error source is the randomized sketch rather than finite precision. Second, mixed precision and naive float16 have finite-precision errors of the same order. This suggests that, for these small synthetic experiments, storage quantization and the sketch itself dominate any benefit from higher-precision accumulation. Thus the experiments support the error decomposition, but they do not by themselves show a strong advantage of mixed precision over naive float16.

Configuration	s	E_{sk}	E_{fp}	$E_{\mathrm{sk}}/E_{\mathrm{fp}}$
`full_fp32`	8	6.23e+02	9.92e-05	6.3e+06
	16	4.73e+02	9.79e-05	4.8e+06
	24	3.49e+02	7.86e-05	4.4e+06
	32	3.10e+02	7.42e-05	4.2e+06
`full_fp16`	8	6.23e+02	3.70e-01	1.7e+03
	16	4.73e+02	2.51e-01	1.9e+03
	24	3.49e+02	2.12e-01	1.6e+03
	32	3.10e+02	1.82e-01	1.7e+03
`mixed_fp16`	8	6.23e+02	3.38e-01	1.8e+03
	16	4.73e+02	2.30e-01	2.1e+03
	24	3.49e+02	1.95e-01	1.8e+03
	32	3.10e+02	1.67e-01	1.9e+03

Table 7 / Hilbert–Schmidt errors of the logit matrix ($n = 256$, $d = 64$, averaged over 3 seeds). Sketching error $E_{\mathrm{sk}} := \|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\mathrm{HS}}$; finite-precision error $E_{\mathrm{fp}} := \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\mathrm{HS}}$; ratio $E_{\mathrm{sk}}/E_{\mathrm{fp}}$.

5 Outlook

The present report studies only the feature-sketch model $QK^T \approx QSS^TK^T$. The natural extensions are Nyström-type approximations and random-feature approximations of the softmax kernel, which would connect the project more directly with the efficient-attention literature [5] [6]. Another direction is to replace the Gaussian sketch by a structured random matrix (e.g., a subsampled randomized Hadamard transform), which would yield the same probabilistic guarantees with lower computational cost.

References

Back-references to the pages where the publication was cited are given by •.

[1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. In: *Advances in Neural Information Processing Systems 30*. **2017**.
arXiv: 1706.03762 [cs.CL] 1

[2] Nathan Halko, Per-Gunnar Martinsson, and Joel A. Tropp. Finding Structure with Randomness: Probabilistic Algorithms for Constructing Approximate Matrix Decompositions. *SIAM Review*, **2011**.
DOI: 10.1137/090771806 1, 4

[3] Michael W. Mahoney. Lecture Notes on Randomized Linear Algebra. **2016**.
arXiv: 1608.04481 [cs.DS] 1

[4] Sinong Wang, Belinda Z. Li, Madian Khabsa, Han Fang, and Hao Ma. Linformer: Self-Attention with Linear Complexity. **2020**.
arXiv: 2006.04768 [cs.LG] 1

[5] Yunyang Xiong, Zhanpeng Zeng, Rudrasis Chakraborty, Mingxing Tan, Glenn Fung, Yin Li, and Vikas Singh. Nyströmformer: A Nyström-Based Algorithm for Approximating Self-Attention. **2021**.
arXiv: 2102.03902 [cs.CL] 1, 7

[6] Krzysztof Choromanski, Valerii Likhosherstov, David Dohan, Xingyou Song, Andreea Gane, Tamas Sarlos, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, David Belanger, Lucy Colwell, and Adrian Weller. Rethinking Attention with Performers. **2021**.
arXiv: 2009.14794 [cs.LG] 1, 7

[7] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, and Hao Wu. Mixed Precision Training. In: *International Conference on Learning Representations*. **2018**. 1

[8] Nicholas J. Higham. Accuracy and Stability of Numerical Algorithms. 2nd ed. SIAM, **2002**.
DOI: 10.1137/1.9780898718027 5